



GENERAL ZOOLOGY

COURSE (100Z)

(2018-2019)

Part II : Histology

Histology

A branch of biology dealing with studying of tissues; in terms of structure, location in the body and their functions.

Tissues

Tissues are group of cells that are similar in structure and perform common or related functions.

Types of the Tissues

In higher animals, their body has 4 types of tissues:

It fills internal space, structural support, storage of energy.

It contracts for specific movements.

connective
tissue

Muscular
Tissue

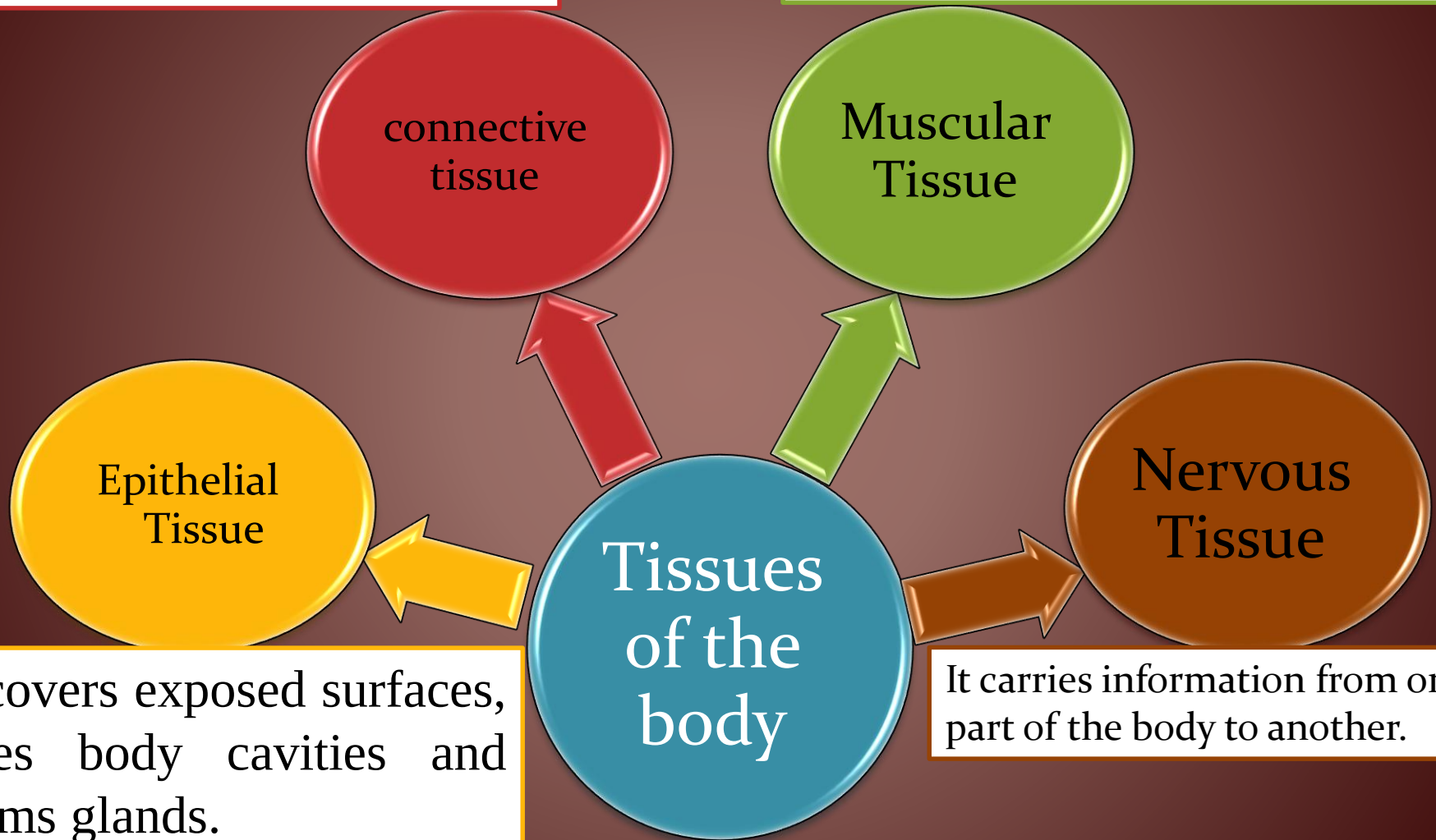
Epithelial
Tissue

Nervous
Tissue

Tissues
of the
body

It covers exposed surfaces,
lines body cavities and
forms glands.

It carries information from one
part of the body to another.



Epithelial
Tissue

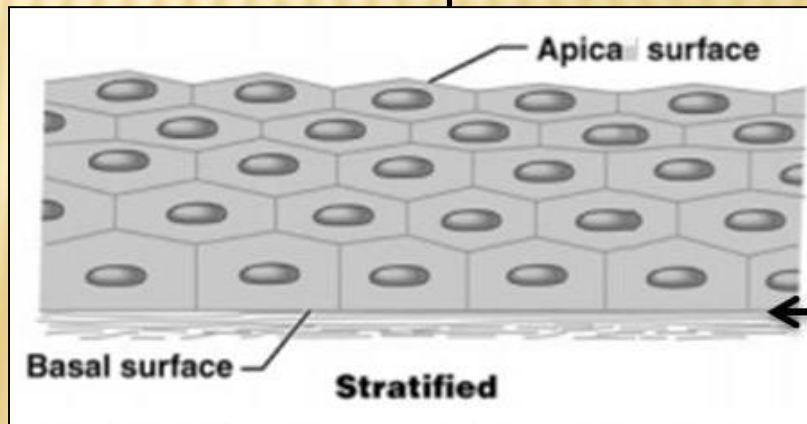
Characteristics of epithelial tissues:

1- Cellularity

Which mean that this type of tissues has little extracellular material (small spaces between cells).

2-Polarity: The epithelial cells have **an exposed surface (apical surface)** and an attached surface-basal surface (attached to underlying tissue). The basal surface of an epithelium is bound to a thin basement membrane.

Fig. Polarity



Characteristics of epithelial tissues:

3- Blood supply:

- is the way by which tissues can receive nutrients from blood through blood vessels.
- Epithelial tissue is avascular tissue (Has no blood vessels).
The blood vessels don't penetrate between epithelial cells (except in endocrine glands).

So how epithelial tissues receive nutrients?

Epithelial cells receive nutrients by diffusion through apical and basal surfaces (from underlying connective tissue).

Characteristics of epithelial tissues:

4- Regeneration (repair):

Epithelial cells damaged or lost at the apical surface are replaced rapidly.

5- Innervated:

Nerve fibers can penetrate between epithelial cells. It is penetrated by the nerve endings making it innervated.

6- Metaplasia التحويل النسيجي:

One type of epithelial tissue may change into another (examples: pseudostratified epithelium of respiratory passages transforms into stratified squamous epithelium on the surface of epiglottis).

Types of epithelial tissue

A- According to number of layers present in the tissue:

There are two types:-

Simple epithelium

composed of one layer of cells.

Stratified epithelium.

composed of two or more layers of cells

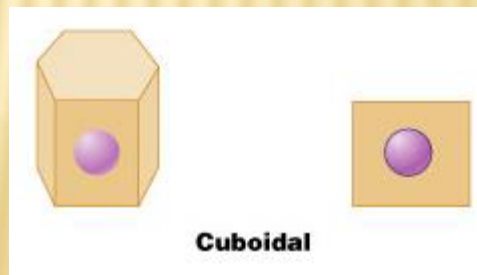
B- According to shape of the cells present in :

There are 3 types of epithelium:

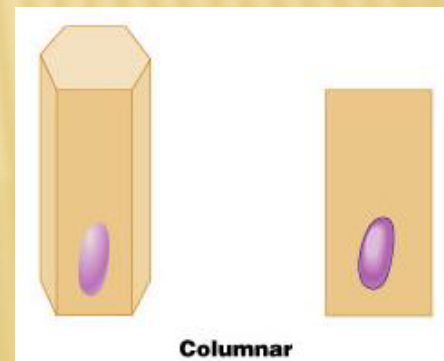
Squamous



Cuboidal:



Columnar



Types of epithelial tissue

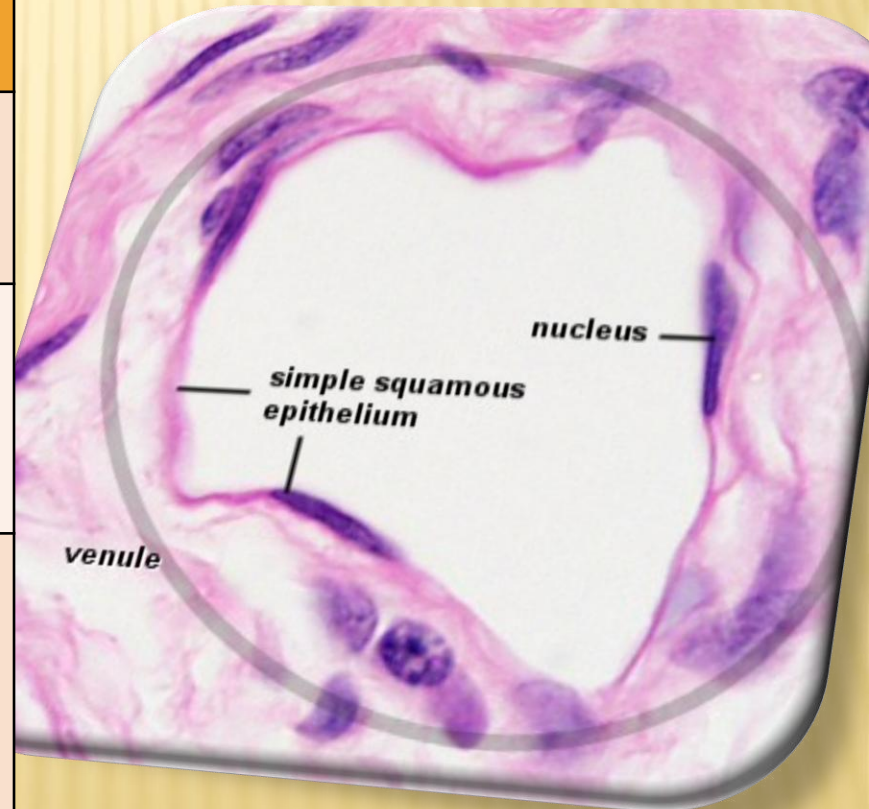
A. Simple epithelium

1) Simple squamous cells

squamous cells

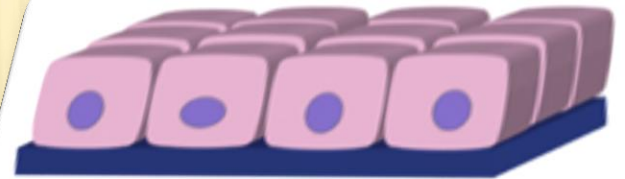


	1. Simple squamous epithelium:
Description	composed of one layer of cells, squamous in shape (Flat cells with flattened nuclei)
Location :	•walls of blood vessels •It forms the outer layer of Bowman's capsule of the kidney •It forms serosa around intestine
Functions	•Found where exchange is important □ e.g., Facilitates filtration of urine in kidneys. •ii) Secretes lubricating substances in serosae



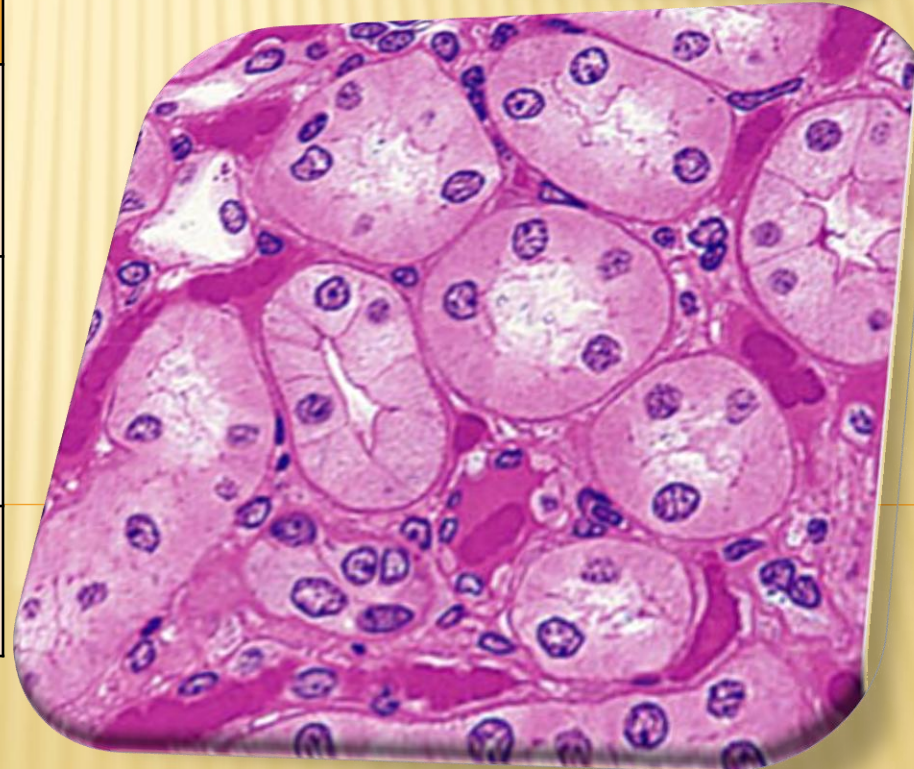
A. Simple epithelium

2) Simple Cuboidal epithelium



Simple Cuboidal

	2- Simple Cuboidal Epithelium
Description	Single layer of cube-like cells with large, spherical central nuclei.
Location:	•Kidney tubules, •ducts and secretory portions of small glands •(ovary surface).
Functions	Secretion and absorption



A. Simple epithelium

3) Simple Columnar epithelium

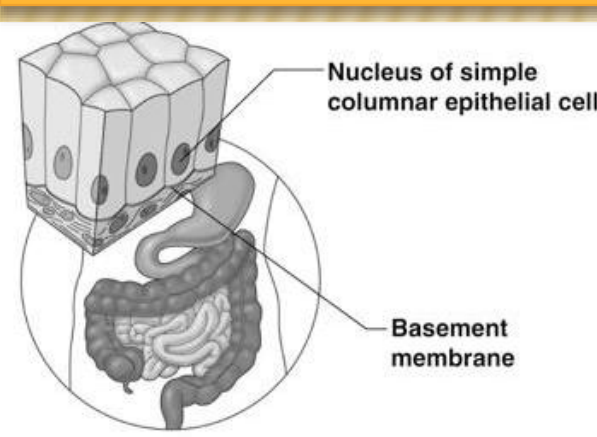
Description: Single layer of column-shaped (rectangular) cells with oval nuclei

According to presence or absence of cilia on simple columnar epithelium, there are two types:

Nonciliated

simple columnar epithelium:
(it has no cilia on the cell surface.)

Function : Absorption; secretion of mucus, enzymes, and other substances.



Ciliated

simple columnar epithelium:
(it carries cilia on the cell surface)

Function : Ciliated type moves mucus or reproductive cells by ciliary action.



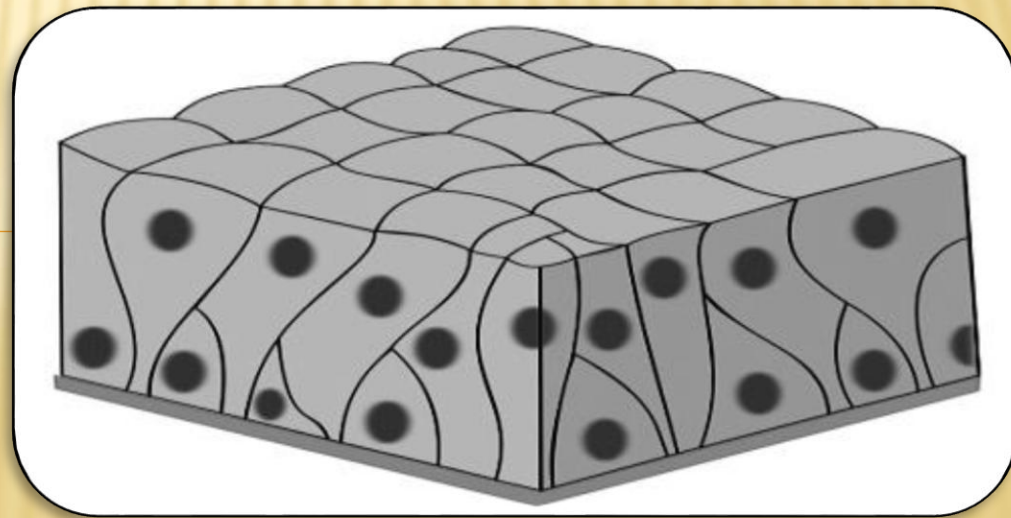
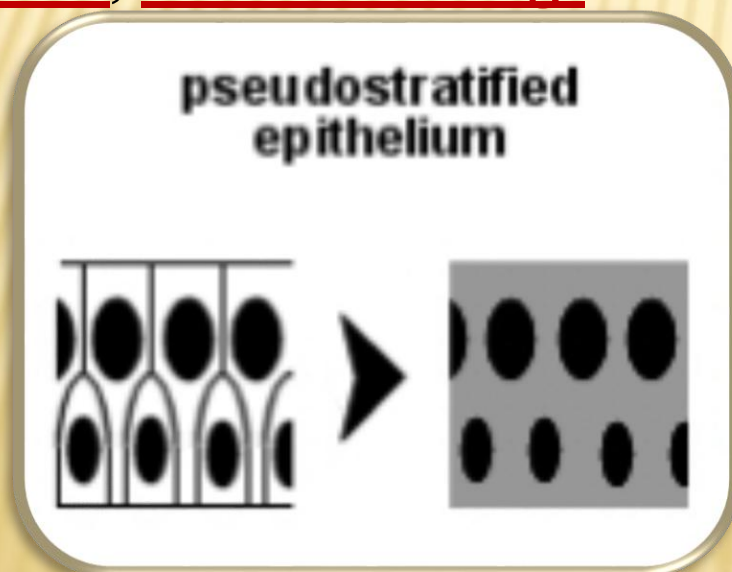
A. Simple epithelium

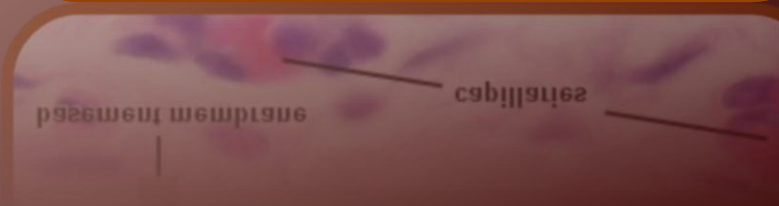
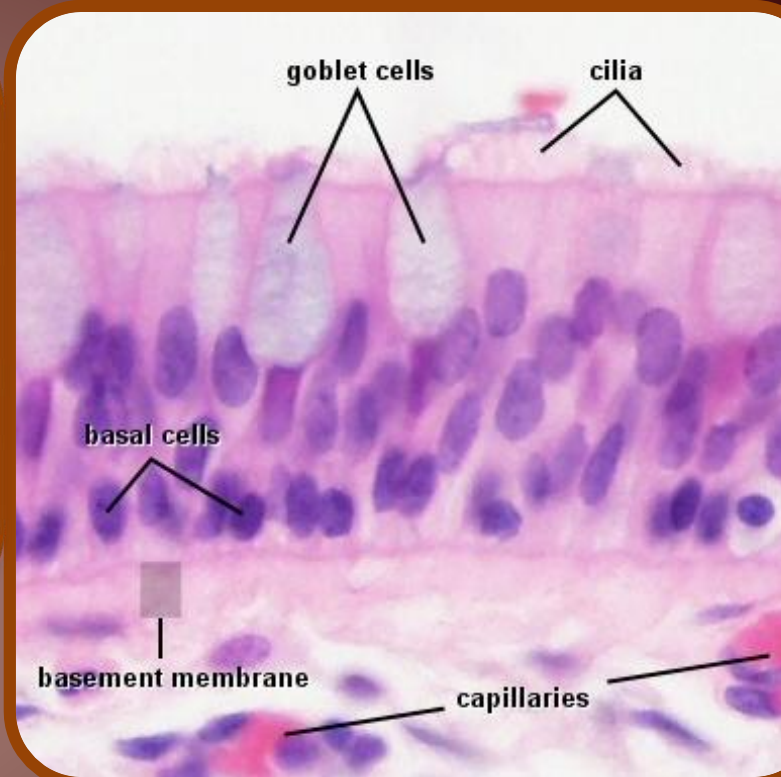
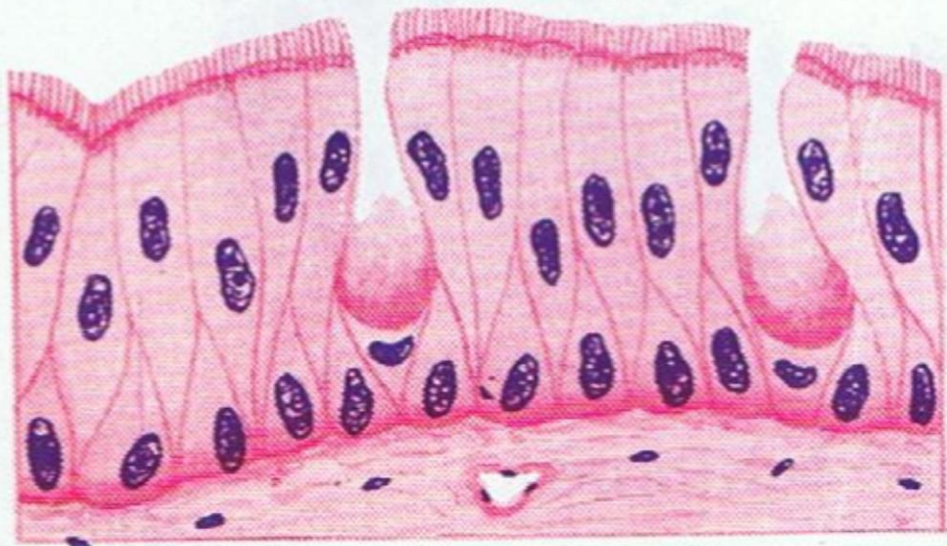
4- Pseudostratified columnar epithelium

Description and location: It consists of single layer of columnar cells but nuclei situated at different levels in the cell giving false appearance of multiple layers.

All cells are in contact with the basement membrane, but not all cells reach the apical surface.

This type is present in the respiratory passages – nasal cavity, larynx, trachea, bronchi of lungs.



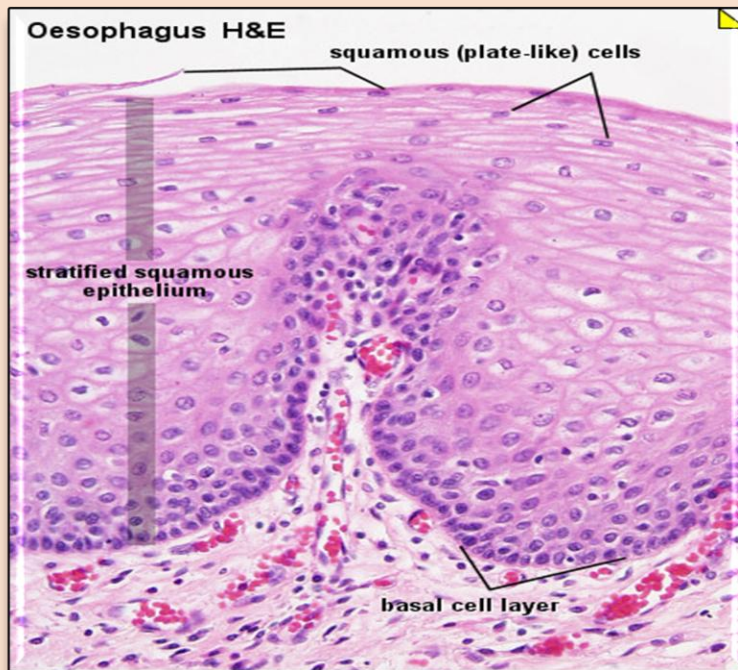


B. Stratified epithelium

1) Stratified Squamous Epithelium

- consists of multiple layers, with squamous cells at the apical surface.
- There are two types of stratified squamous epithelium :

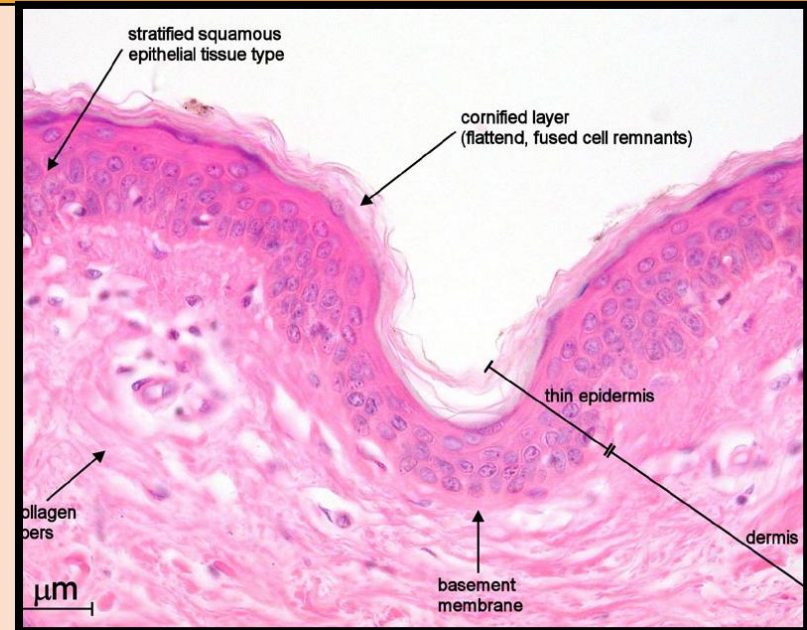
Nonkeratinized stratified squamous epithelium



It has no keratin.

found in the **lining of oesophagus**

Keratinized stratified squamous epithelium (in Skin).



where the most superficial (outer) layers of cells are dead and contains the fibrous protein keratin.

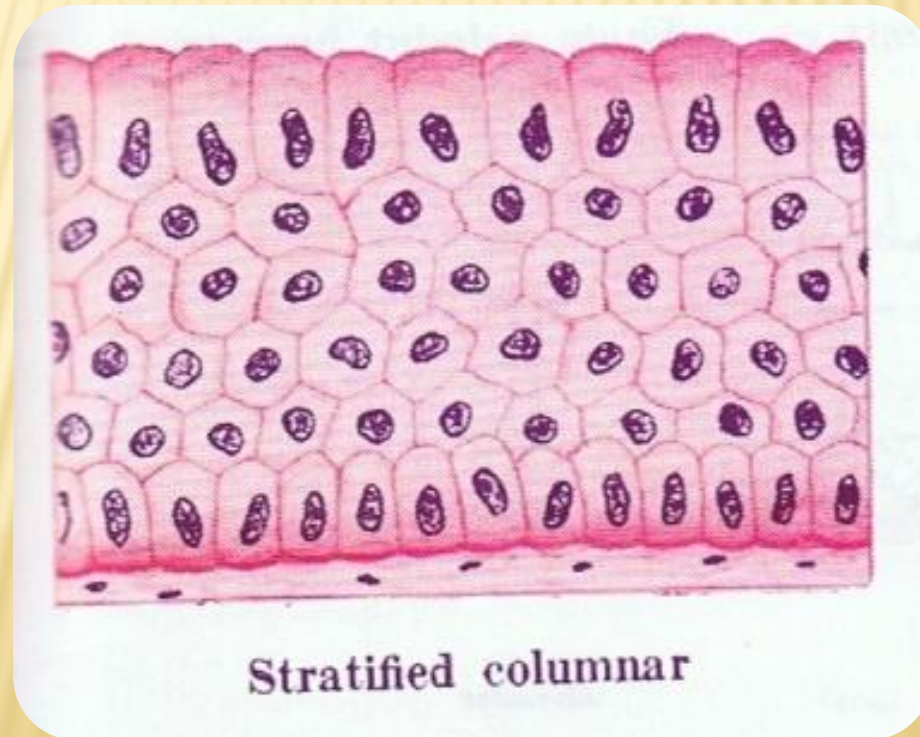
covering the skin

B. Stratified epithelium

2) Stratified columnar Epithelium

Description and location:

Two or more layers of cells, columnar cells form the upper layer
(two-layered in ductus deferens)
(more-layered male urethra)



B. Stratified epithelium

3) Transitional :

Description and location:

Epithelium change the shape of cells and number of layers according to wall conditions of urinary passages – distension or contraction.

In the urinary bladder:

this tissue is extending in size when the bladder is empty.

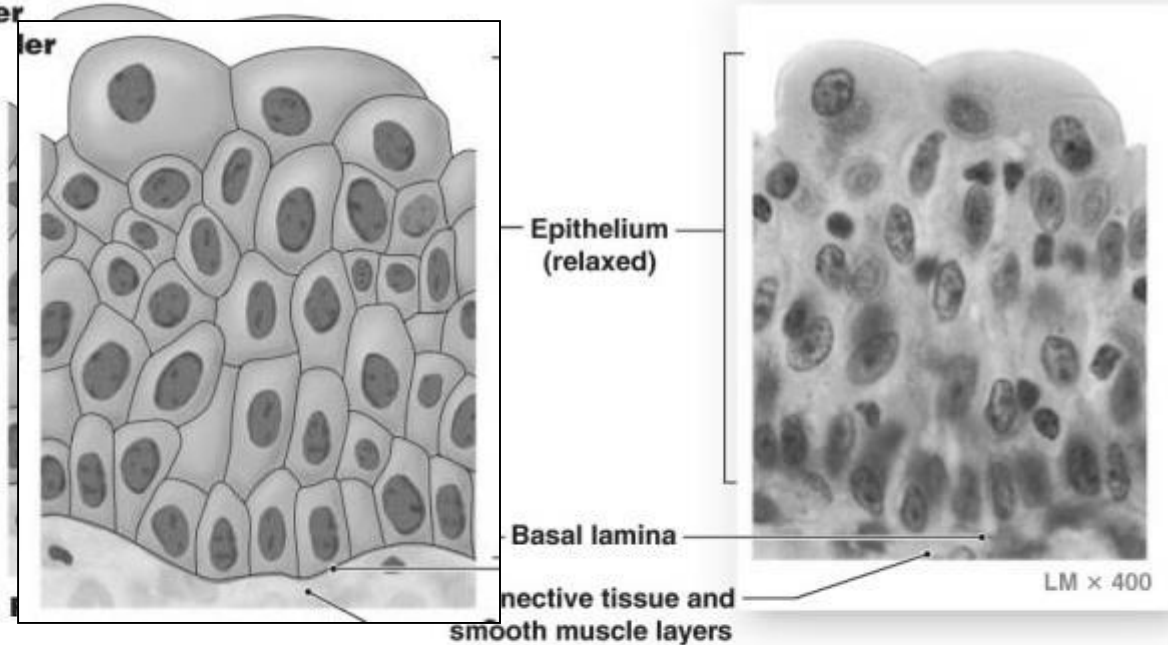
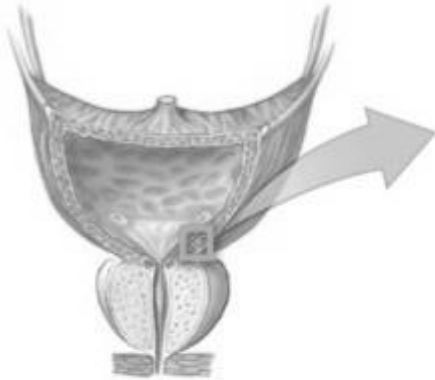
In full bladder:

urine press on the tissue making reduction in size of the tissue.

The transitional epithelium in an empty and a full urinary bladder

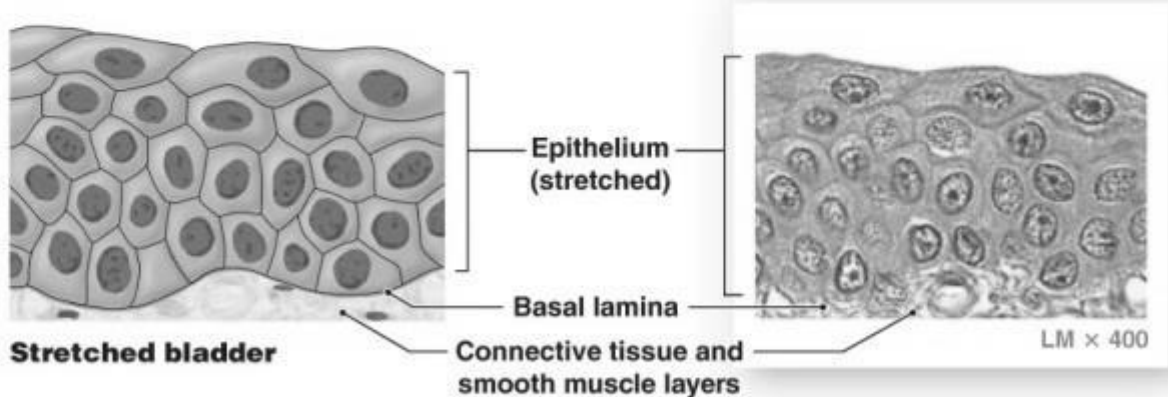
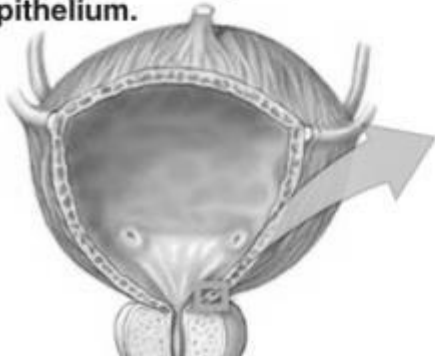
Epithelium in a Relaxed Bladder

In an empty urinary bladder, the superficial cells are cuboidal with a dome-shaped surface.



Epithelium in a Stretched Bladder

When the urinary bladder is full, the volume of urine has stretched the lining to such a degree that the epithelium appears flattened, and more like a stratified squamous epithelium.



3) Glandular epithelium

GLANDS are cells or aggregations of cells whose function is secretion according to the presence or absence of their ducts can be divided into :

- **Endocrine glands** release their secretory product (typically hormones) into the spaces between the secretory cells (extracellular space) from which it enters the bloodstream.
- **Exocrine glands** release the secretory product via a system of ducts that opens upon one of the surfaces of the body which are in contact with the external world (skin, gastrointestinal tract etc.).

Classification of Exocrine Glands:

Exocrine glands may be classified according to cell number, and/or the shape and branching pattern of their secretory portions and ducts.

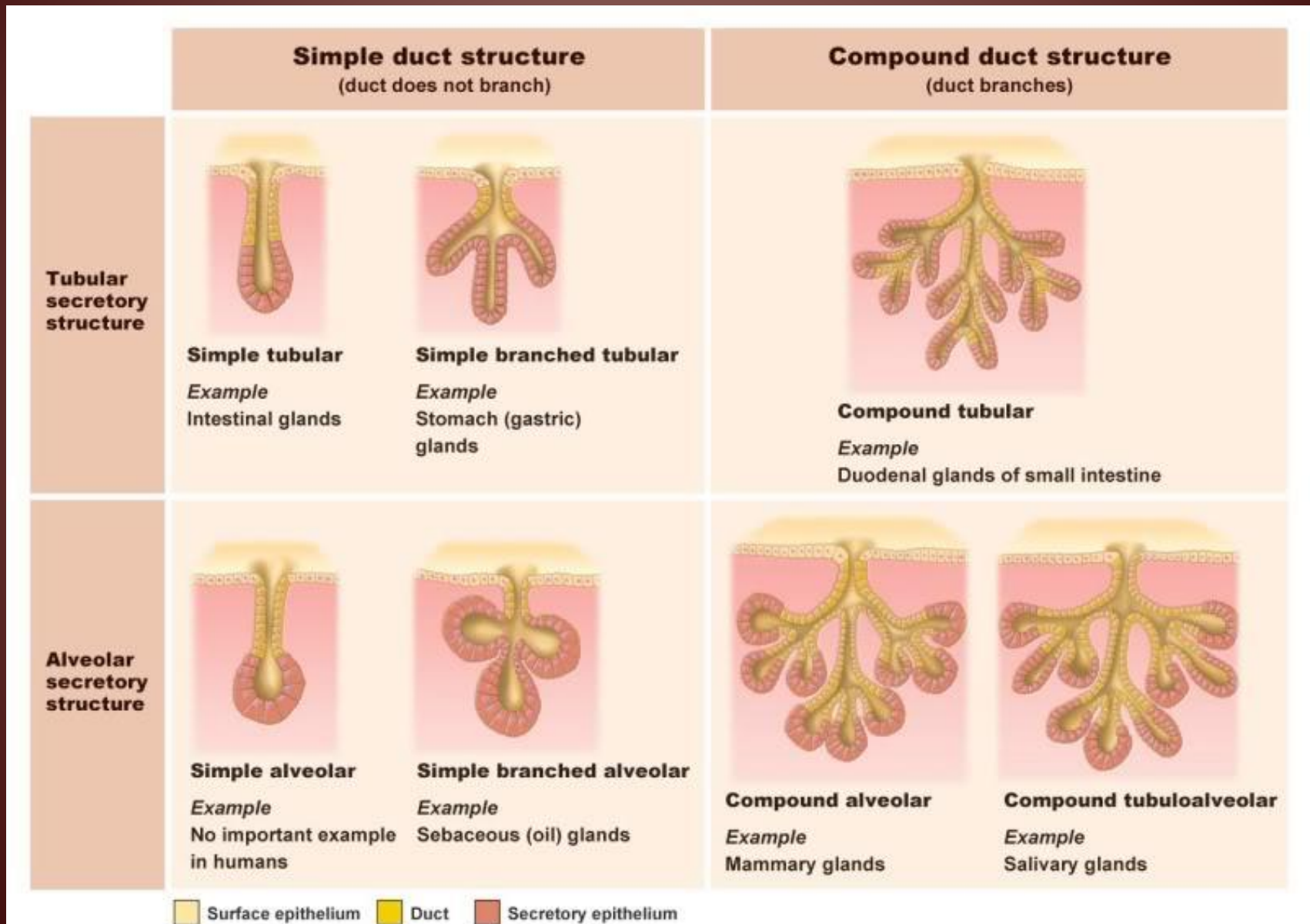
➤ according to cell number

■ Unicellular Glands :

consist of a single secretory cell. In mammals the only example of unicellular exocrine glands are ***goblet cells***, which occur in the epithelium of many mucous membranes. Goblet cells secrete the glycoprotein mucin, which by the uptake of water is converted into a slimy substance, mucus.

■ Multicellular glands .

- According to the shape and branching pattern of their secretory portions and ducts multicellular glands classified as the following.



Connective tissue

Connective

Origin: **tissues:**

All types of connective tissue originate from the mesoderm.

Types:

there are three different types of connective tissue:

- 1- C. T. proper** it may be loose (Areolar, Adipose, Reticular) or may be dense (Dense regular, Dense irregular and elastic).
- 2- Skeletal C. T.** (it constitutes both bone and cartilage)
- 3- Vascular C.T.** (Blood).

Components :

Typical connective tissue consists of:

- cells embedded in an extracellular matrix (ECM).
- ECM (Extracellular matrix) has a ground substance,
- fibers and tissue fluids.

Components of connective

- **Cells of connective tissue: tissues:**

- 1- **Fibrocytes**: this type of cells is present mainly in proper connective tissue.
- 2- **Chondrocytes and osteocytes**: are present mainly in cartilages and bones (skeletal C.T.)
- 3- **Blood cells**: which are circulating in the plasma of the vascular connective tissue.

- **Fibers of connective tissue**

- 1- **Collagen Fibers** (**white fibers**):

Composed of the **protein “collagen”**.

- 2- **Elastic Fibers** (**Yellow fibers**):

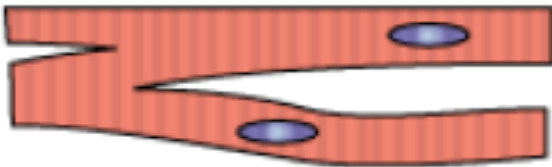
Composed of **elastin protein**, less strength than collagen.

- 3- **Reticular fibers**:

Fine collagenous fibers; branched and form networks; surround small blood vessels and support tissue of organs.

Functions of C. T.

1. Binding and support (ex. Tissues surrounding intestine, bone)
- 2 .Protection (ex. Blood)
- 3 .Thermal Insulation (Adipose C. T.)
4. Transportation (ex. Blood)

Skeletal		Voluntary	Striated	Multinucleated	Non-branched
Cardiac		Involuntary	Striated	Single nucleus	Branched
Smooth		Involuntary	Nonstriated	Single nucleus	Tapered

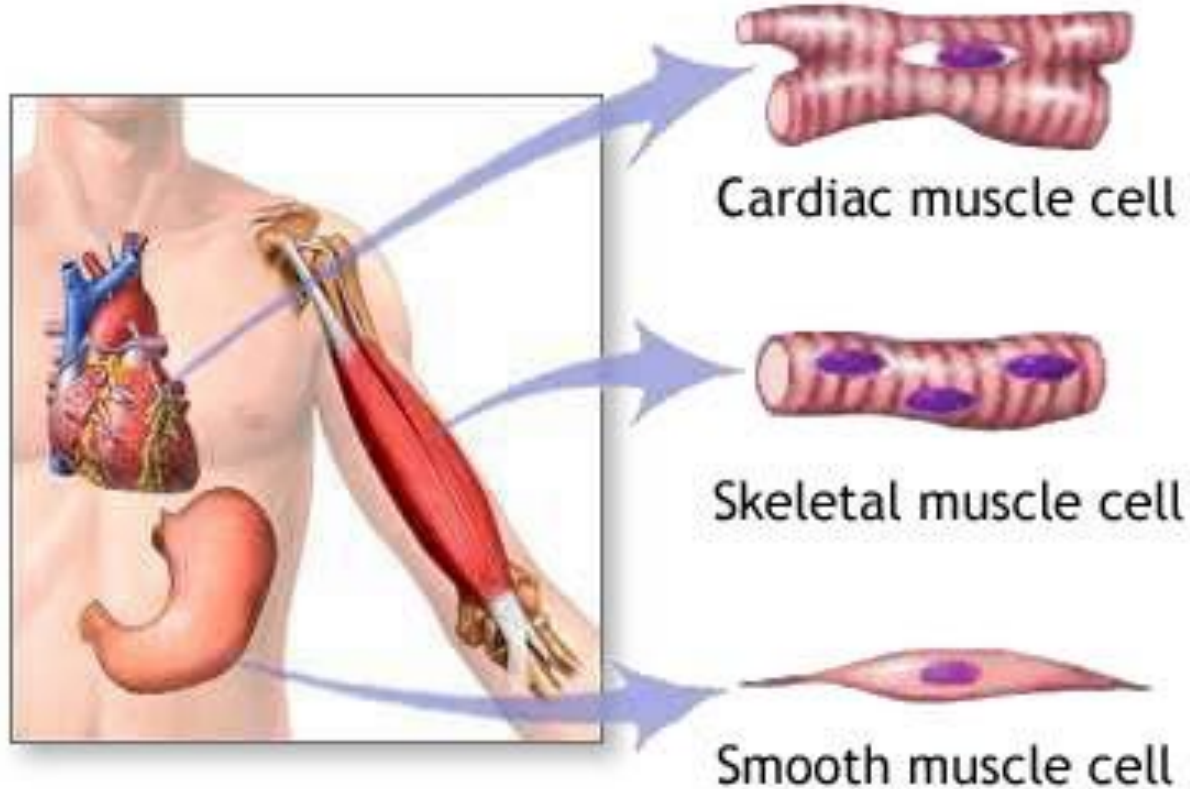
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Fig. Differences between different muscles

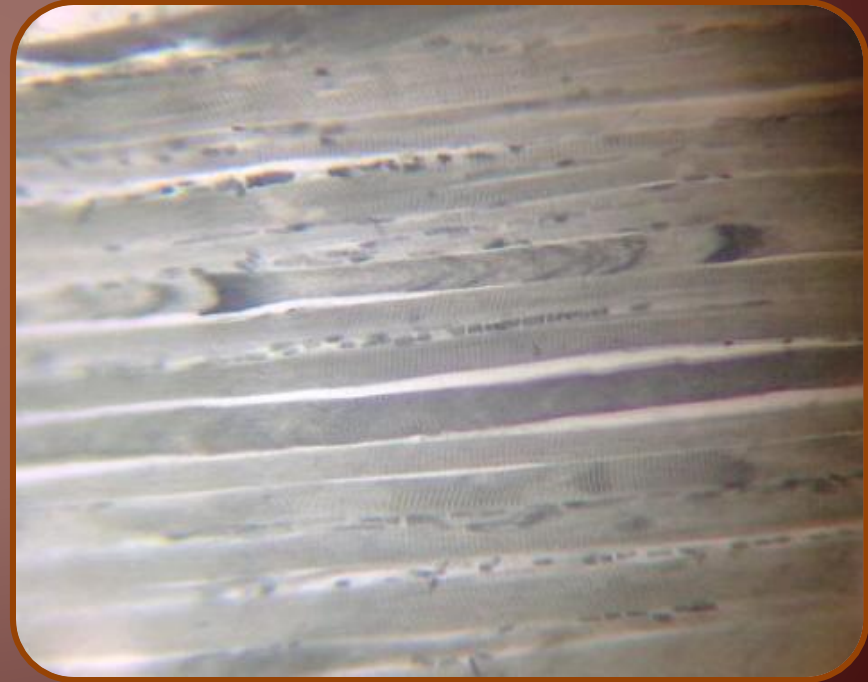
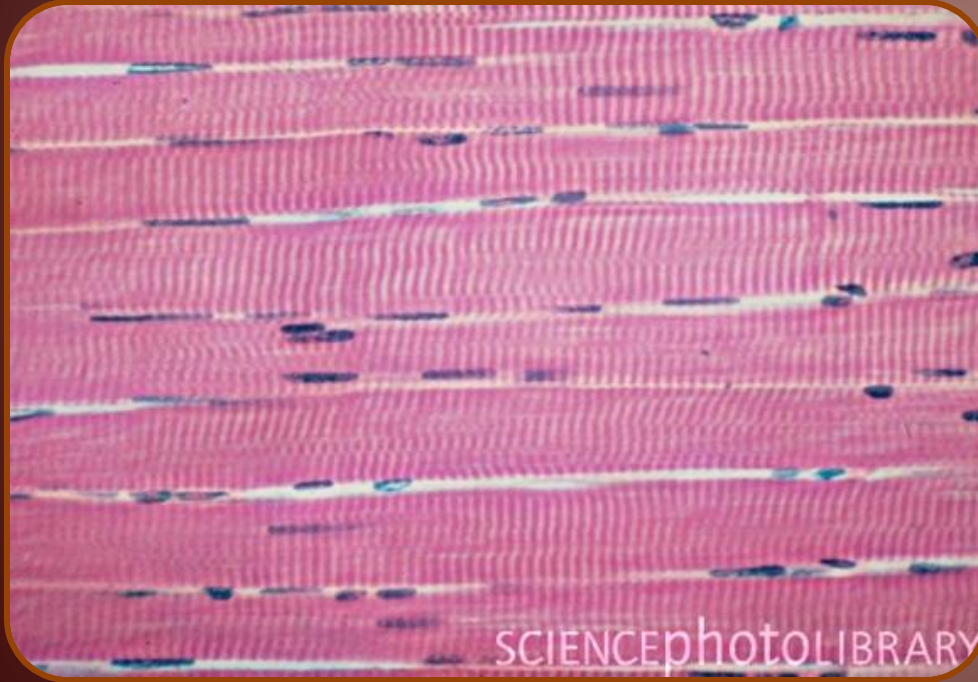
Muscular Tissue

Mostly animals has three types of muscles :

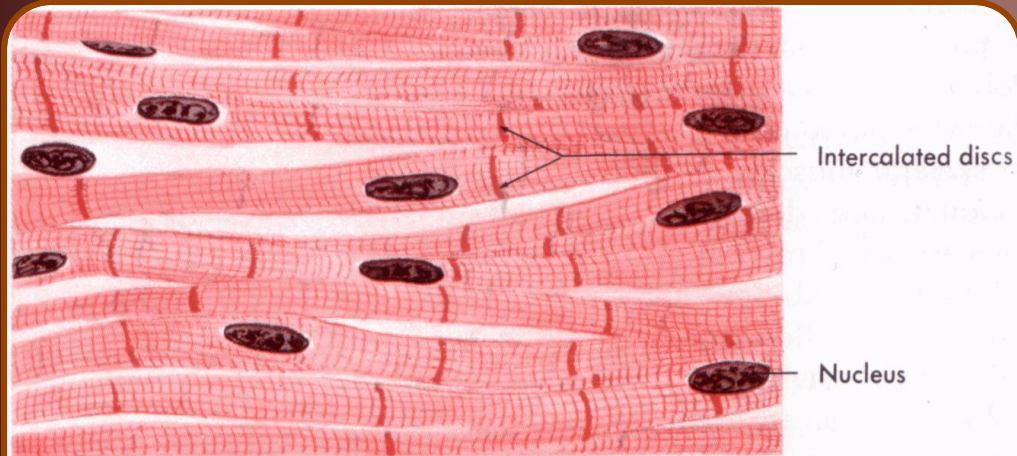
- Cardiac muscles - Smooth muscles - Skeletal muscles



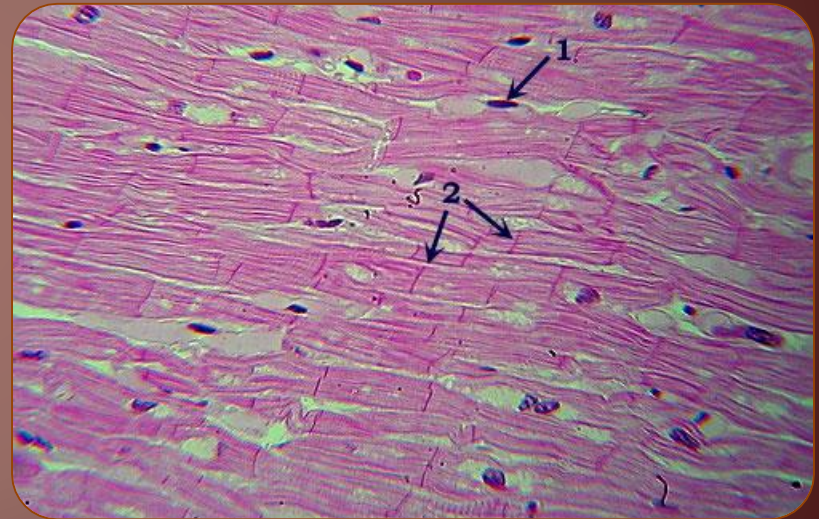
Skeletal Muscles



Cardiac Muscles



Cardiac or striated involuntary muscle tissue.



Cardiac or striated involuntary muscle tissue.